



BARISTA

FUNDAMENTALS

WHAT IS ESPRESSO?

Espresso is simply a brewing method, where a small amount of hot water is forced through finely ground coffee under high pressure.

Thus...

ANY COFFEE CAN BE ESPRESSO!

If a bag of coffee is labeled with 'espresso blend' or 'espresso beans', this mainly indicates that the roaster recommends it to be brewed as espresso. But it could be delicious as a French Press too!

There are a myriad of techniques that can be used to produce great espresso. In this guide and class, we humbly present best practices for you to take home and build upon.

TOOLS OF THE TRADE

COFFEE GRINDER

A. HOPPER

where the beans are stored/loaded before grinding

B. BEAN GATE

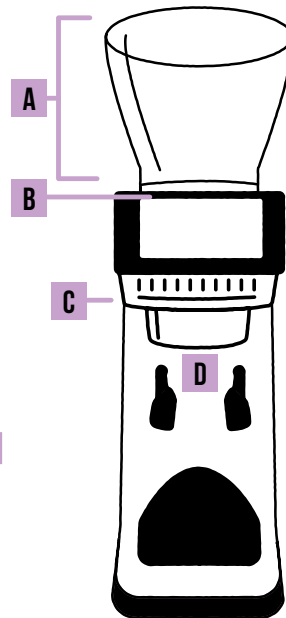
when open allows beans to fall through, when closed prevents any beans from being ground

C. GRIND ADJUSTMENT RING/COLLAR

for making your espresso coarser or finer

D. CHUTE/DOSER

where the ground espresso comes out



There are many types of home espresso machines out there. The key is to get one that can heat & keep the water at a consistent temperature for your shots and has good pressure for steaming.

A. PORTAFILTER

where you dose and pack the coffee grounds

B. GROUPHEAD

where you lock in the portafilter

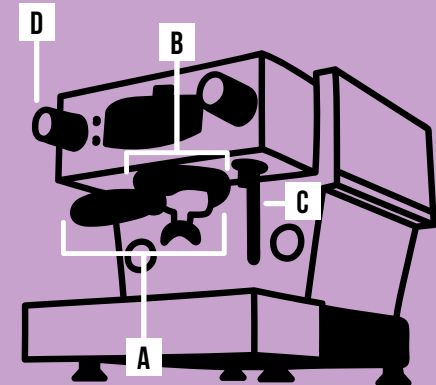
C. STEAM WAND

for heating, texturizing, and steaming milk

D. PRESSURE GAUGE

displays the pressure of your coffee/steam boiler, usually most important for troubleshooting purposes

ESPRESSO MACHINE

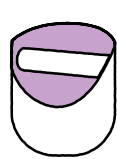


ACCESSORIES



TAMP

to press and compact coffee grounds



KNOCK BOX

for knocking used espresso "pucks" out of your portafilter



SCALE

for best accuracy in weighing ground coffee and espresso

PULLING A SHOT



1 Remove portafilter and flush the grouphead

Knock out any grounds from the previous shot, if necessary.

2 Wipe the portafilter

Make sure it's clean and dry.

3 Grind coffee dose into portafilter

Try to have grounds fall in an even mound in the center of the portafilter.

4 Weigh dose on scale to confirm correct amount of coffee

Essential for maintaining a proper recipe and coffee-to-water ratio.

5 Level/distribute

Disperse and settle the bed of grounds to ensure even extraction.

6 Tamp

Compact the espresso grounds, making sure to stay level.

7 Check your puck surface

If there are any cracks or big divots, knock out the coffee and start again.

8 Flush the grouphead again

Flush 1-2oz of water (no more) to maintain temperature.

9 Load portafilter

Gently re-insert the portafilter into its grouphead. Roughly slamming the portafilter back into the machine will create cracks and channels in the bed of coffee that will yield a poor extraction.

10 Pull shots/begin extraction

Pull them into the rinsed shot glasses, or a mug.

11 Monitor/diagnose shots

Proper timing and use of visual cues are essential to maintaining quality.

UNDERSTANDING EXTRACTION VS STRENGTH

WHAT IS EXTRACTION? WHAT IS STRENGTH?

EXTRACTION

is the act of pulling soluble compounds out of coffee grounds by using water.

STRENGTH

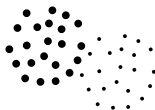
is the amount of solubles present in the finished cup, also called Total Dissolved Solids (TDS).

We control the extraction by manipulating variables that impact the water's access to soluble material, creating and affecting coffee flavor.



RATIO

This is the overall brew strength of your shot. How much coffee is going into the portafilter in relation to how much water we are running through it.



GRIND PARTICLE SIZE

The coarseness or fineness of your coffee. This directly relates to the rate at which the soluble material extracts from the coffee.

Finer grind = slower shots
Coarser grind = faster shots



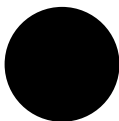
BREW TIME

How long we are allowing the water to extract soluble material from the puck of coffee.

In order to make consistently delicious espresso shots, it is extremely helpful to use a recipe, or Ratio — this will keep your **STRENGTH** constant so that you can focus on manipulating the **EXTRACTION**.

We will use scales to weigh the amount of ground coffee going into the portafilter, and to weigh the amount of liquid espresso coming out of the grouphead. A popular ratio that complements a lot of coffee is 1:2, or 1 part ground coffee IN, to 2 parts espresso OUT. Depending on the size of your portafilter, this might be something like 17 grams of ground coffee IN, to 34 grams of espresso OUT.

QUICK HITS



GROUND COFFEE



Example: 17g ground coffee | 34g espresso

DIALING IN AND ADJUSTING YOUR GRIND SETTING

- When a barista says they are 'dialing in,' it means they are making adjustments to their espresso grinder, in order to make the shots extract better.
- First, verify your dose weight (Ratio) is what you intended. If shots are running too slow, (generally reaching their total weight 6-10 seconds slower than your intended recipe) that is referred to as **over-extracted**.
- If your shots are running too fast, (generally reaching their total weight 6-10 seconds faster than your intended recipe) this is referred to as **under-extracted**.
- Adjust your grind size coarser or finer to correct the problem. Purge one portafilter of coffee after each grind adjustment to clear the chamber of any grounds from the previous setting.
- Now you are ready to pull your next shot, to confirm if you have brought your espresso back to an ideal extraction!

- Repeat the grind adjustment as needed, until you are **DIALED IN!**

OVER-EXTRACTED

The water has pulled too much soluble material from the bed of coffee. These shots often taste bitter, with a lingering unpleasant finish.

UNDER-EXTRACTED

The water has not spent enough time with the bed of coffee and has not dissolved enough soluble material. These shots often have a thin mouthfeel, with a sour or papery taste.

UNDERSTANDING GRIND SETTINGS

If you have a hard time recalling which way to adjust your grinder when dialing in, use this example:

IMAGINE YOU HAVE TWO BEACH BUCKETS.



SAND

PEBBLES

One is packed full of beach sand, and the other is packed full of pebbles. If you were to pour water over the top of the two buckets, which one would the water flow through faster?

The sand is like your fine grind, and the pebbles are like your coarse grind.

EVERYTHING YOU NEVER THOUGHT YOU'D NEED TO KNOW ABOUT MILK

- Milk can be a great companion to espresso because the sugars and fats complement the bitterness and dial down the intensity.
- Whole milk is the best for steaming practice, and it is actually only about 4% milkfat. The easiest non-dairy for steaming is usually a “Barista” version of Oat Milk.
- Properly steamed milk will be naturally sweet with a smooth, even texture. The bubbles should be almost invisible to the eye; this is called “microfoam”.
- It should have a glossy appearance and viscosity similar to a can of white house paint.
- Dairy milk burns at 165°F. The optimal temperature range to steam milk within is 140°F-150°F.
- Latte art is ONLY possible if you've mastered the process of steaming milk.



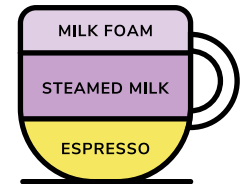
COMMON ESPRESSO DRINKS



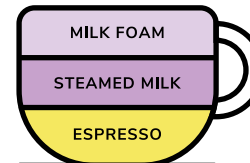
ESPRESSO
2oz



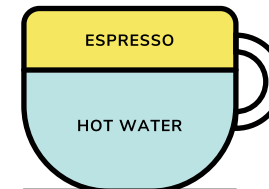
MACCHIATO
3oz



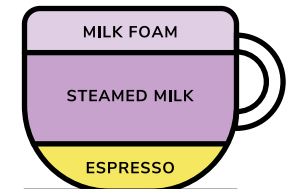
CORTADO
4oz



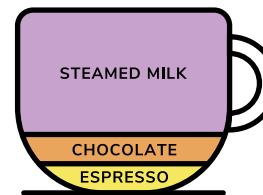
CAPPUCCINO
8oz



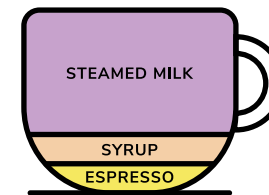
AMERICANO
12oz



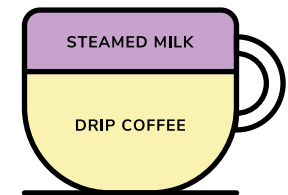
LATTE
12oz



MOCHA
12oz



FLAVORED LATTE
12oz



CAFE AU LAIT
12oz

HOW TO STEAM MILK

STEP-BY-STEP



- 1** Pour milk into a pitcher, about 1/4 inch below the spout.
- 2** Purge the steam wand, and place it into position — imagine it as aiming toward your toes.
- 3** Submerge the tip of the steam wand into the center of the milk, just barely covering the end.
- 4** Hold the pitcher handle with one hand. The other hand will activate the steam wand, and then immediately come down to rest against the side of the pitcher to monitor the increase in temperature.
- 5** When you're ready, turn on the steam wand and immediately begin the stretching/aerating phase. It should sound like a soft hissing or paper-tearing noise.
- 6** As soon as the milk's temperature becomes 100°F, raise the pitcher up about 1/4 inch and give it a slight tilt.
- 7** At this point the hissing noise should cease and you should see a circular motion, or "vortex" begin to happen.
- 8** When the pitcher becomes too hot to comfortably touch with your free hand, reach up and turn the steam wand off.
- 9** Lower the pitcher and immediately wipe the milk residue off your steam wand, with a downward twisting motion.
- 10** Give the pitcher a light tap against the counter to knock out any large bubbles. Then swirl the pitcher to mix the milk. You should see it turn glossy, looking like "wet paint" or "melted ice cream"
- 11** Optional: Use an instant read thermometer to confirm the temperature is between 140-150 degrees.

HOW TO POUR LATTE ART

STEP-BY-STEP



1 Pour your espresso into the cup and cradle it in the palm of your hand, tilting it towards the pitcher in your other hand.

2 Start your pour up high (about 3-4" above the surface of the espresso) to create the canvas for your art. Ideally the milk should dive below the surface of the espresso, preserving the rich brown color as much as possible.

3 When the cup is about 2/3 full, quickly bring the tip of your pitcher down and centered, as close to the liquid in the cup as possible. At this point you should see a crisp white dot begin to appear, because the milk is gliding across the surface (if you do not see white appear, you are not close enough).

4 Once you have formed the dot, hold steady and it will continue to grow as your cup fills.

5 The final step is called the "pull through." Once the cup is full, lift your pitcher back to the starting position and pull it directly across the center of your dot. When done correctly, this will appear as if an invisible thread is pulling through your dot, turning it into a heart. The pull through movement should mimic a helicopter taking flight, rather than an airplane.

REMEMBER TO PRACTICE, AND DON'T BE TOO HARD ON YOURSELF—LATTE ART IS FUN TO MASTER, BUT IT DOESN'T TASTE LIKE ANYTHING!

CLEANING YOUR MACHINE

THERE IS NOTHING WORSE THAN HAVING YOUR EFFORTS THWARTED BY DIRTY EQUIPMENT.

CLEANER EQUIPMENT
=
BETTER TASTING COFFEE

Espresso Machine

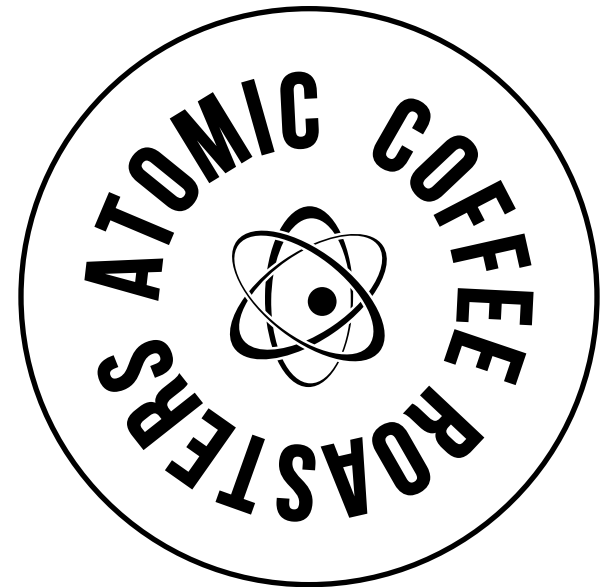
1. Scrub grouphead gasket with a Pallo tool or wipe grouphead with a cloth.
2. Remove the portafilter basket and dispersion screen from the machine.
3. Backflush with Cafiza and blind disk.
4. Soak portafilters, baskets and dispersion screens in Cafiza solution.
5. Remove and rinse drip and drain tray.
6. Reassemble machine and surface clean.

Grinder

1. Close grinder's bean gate and purge residual coffee through.
2. If you are not planning to use the machine for a while, remove grinder's hopper and empty beans into an airtight container for storage.
3. Brush off any grounds from the grinder or counter with a towel.

RECOMMENDED URNEX PRODUCTS:

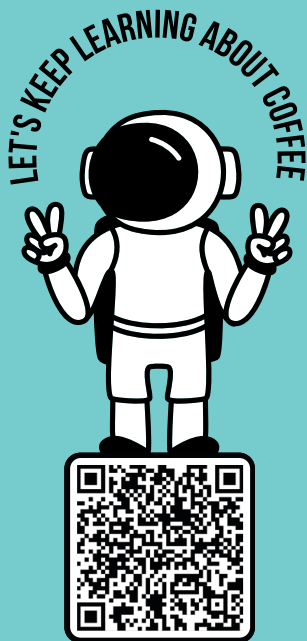
Cafiza, Rinza, Grindz



**THANKS FOR
JOINING US!**

NOTES

NOTES



DRINKATOMIC.COM

ATOMIC COFFEE ROASTERS